# Resolving Project Intent into a Kubernetes Deployment Model

Project Task 2: Model Transformations
Project Type: Code Generation
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# 1 Purpose of the Project

## 1.1 Goals

The project converts short deployment documents into Kubernetes files for an existing small cluster. Each project repository holds one `Project Intent` document. This document describes its `Application`s, including their `Process`es, memory and CPU limits, ports, dependencies, and external routes. One `Platform Intent` document defines shared policies and resources, including cluster entry points, health-check timings, and backup policies. Task 1 defined two Ecore metamodels: a source metamodel for the authored documents and a target metamodel for the `Resolved Deployment`. The target model records the values derived for each `Application` and the paths of the files to be generated.

Task 2 implements the transformation to `Resolved Deployment`. It combines all authored models with recorded cluster facts, image digests, and other pinned inputs, producing one resolved model per `Project`. For example, an image name becomes a repository address pinned to a digest, and a dependency declaration becomes a host-and-port combination. Task 3 then generates files from these resolved values.

| Term | Meaning |
|---|---|
| `Process` | One program of an `Application`, running in its own pods. |
| `Dependency edge` | A declaration by one `Process` that it calls a named port of another `Application`. |
| `Pinned input set` | The read-only inputs besides authored documents: node contract (machines and capacity), images lock (image digests), `ClusterState` snapshot, migration proofs, and configuration-file contents. |
| `Asset` | A configuration file mounted into a `Process`. |
| `Exposure` | An external address and routes to an `Application`'s ports. |
| `Switchover` | How a new version replaces the old: blue/green (both versions run simultaneously, and traffic switches to the new once it is healthy), rolling (gradual replacement), or stop-start (old stops before new starts). |
| `ReleaseGate` | The resolved configuration for the service that blocks a blue/green switch until the `Application`'s `Process`es are healthy, preventing traffic from reaching an unhealthy version. |
| `ReconcileUnit` | A group of resources that Flux (the GitOps operator) reconciles together as a unit, with dependencies on other such groups. |
| `Entry point` | A platform listener (class `Tier` in the source metamodel) that admits specified audiences and forwards routes through a proxy. |
| `Revision` | A release identifier computed by hashing an `Application`'s resolved configuration. |
| `Estate` | All `Project`s, the `Platform Intent`, and the `Pinned input set` read by one run. |
| `Flattened model` | `Project Intent` with every shared declaration copied down onto the `Process`es that reference it (class `EffectiveProject`), preserving declaration precedence. |

Table 1: Project Terms Used in This Report

**Constraints and Design Approach.** Task 2 requires ATL or QVT-Operational. The transformation is deterministic: the same pinned inputs always produce the same resolved model, with no reads of live cluster state. Every value the Task 3 templates need is recorded in the resolved model itself, so template tests cannot pass while a template makes an unchecked deployment decision. Where a value cannot be derived, the run aborts with an error message rather than fabricating a default: a missing or fabricated value could otherwise produce a valid-looking file that deploys the wrong configuration. Conformance to the specification's oracle files is verified by byte-exact comparison of resolved values, dependency edges, generated files, JSON, and release digests (Section 4).

## 1.2 Role of the Transformations

Figure 1 shows the pipeline. Xtext converts each document to a source model, and the Task 1 OCL constraints reject invalid ones. Three QVT-Operational transformations (QVTo) then run in sequence.

The first transformation, the flattening, copies declarations made once for an entire `Project` or `Application` down onto each `Process`. Its output is the flattened model, which still conforms to the source metamodel. Although it was implemented in Task 1, it was not reported there, so Section 3.1 covers it here. The second transformation, the resolution, reads the flattened models and the `Pinned input set`, the additional read-only inputs, and derives one complete `Resolved Deployment` per `Project`. It is the main transformation and the subject of this report. The third, the final pass, collects Vault policies from all resolved models to plan one job that installs them in Vault, the platform's secret management service. This requires every `Project`'s output, so it runs after the individual resolutions (Section 3.7).

A small Java exporter writes the resolved dependency edges as JSON for comparison against the oracle. The Acceleo templates of Task 3 then read the `Resolved Deployments` and generate the Kubernetes files.

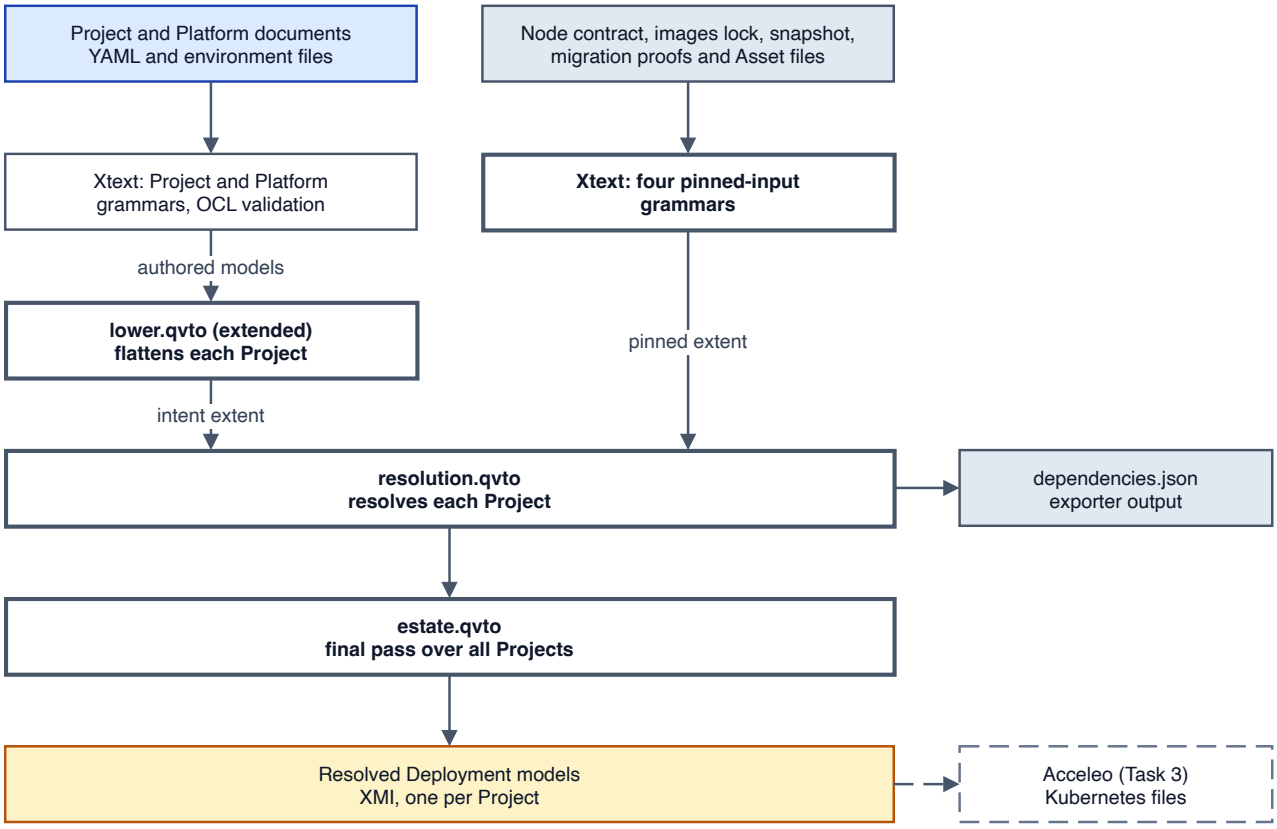



Figure 1: The Pipeline, With What Task 2 Added or Extended in Bold

### 1.3 Refinement of Tasks 0 and 1

Task 0 treated the node contract (the list of machines and their capacity), the images lock (each image pinned to a digest), and the `ClusterState` snapshot as plain inputs, while Task 1 only modelled the two authored documents. The resolution needs these as models, so a third metamodel now describes the `Pinned input set`, with its own Xtext grammar for each document (Appendix A.1). A separate package keeps these supporting inputs distinct from the deployment language used by `Project` authors.

Task 1 described three stages: authored, resolved, and generated text. The flattening now resolves shared declarations before resolution: each `Process` receives the declarations that apply to it from its `Project` and `Application`. Resolution rules therefore need not repeat that merge logic (Section 2.3.1).

Task 1 planned one resolved model for the entire cluster, because resolving dependency edges, routes, and network access requires a cluster-wide view. Every run still reads every `Project`. The output is now one `Resolved Deployment` per `Project`, because each model then stays small and has one oracle to compare with (Section 2.3.2). Each `Application` still has its own `ResolvedApplication` inside that model. The final pass adds the one job that no single `Project`'s resolution can compute (Section 3.7).

Task 1 demonstrated the target metamodel with only the `minimal` example. The other examples needed

additional concepts: migrations, backup service accounts, API access, sidecar containers, and twelve additional file types (see Appendix A for the complete list).

### 1.4 Assumptions and Scope

The transformation assumes four conditions.

- **Checked input.** The OCL constraints check the authored documents before transformation begins. Aborts reject invalid inputs that these constraints do not yet catch (Section 3.8).
- **One cluster per run.** Each execution of the pipeline reads all documents for one cluster, including one `Platform Intent` and one node contract that name the same cluster. All concepts, examples, and oracles derive from that single existing cluster.
- **One reading of the specification.** If several providers satisfy a dependency, the transformation uses the deterministic name order recorded in the oracles. The comparisons check this interpretation, but they do not establish that it is the only valid one.
- **No live cluster access.** The transformation only reads the documents provided to it. Any fact known only to a running cluster, such as which node holds a particular volume, must arrive via the `ClusterState` snapshot captured by another process.

**More Than One Cluster.** A second cluster would need a separate run with its own `Platform Intent`, node contract, and `ClusterState` snapshot. Environment-specific configuration overlays are already selected by cluster name (Section 3.4.3). The model would also need to declare which clusters each `Project` targets and to represent cross-cluster network access with address ranges. Such dependencies could use the external-provider mechanism (Section 3.4.2), already used by the `auth Project` for its mail server. This extension has not been implemented.

Table 2 sets the scope of Task 2.

| In scope | Out of scope |
|---|---|
| The resolution of eight examples: `minimal`, `auth`, `data`, `delivery`, `edge`, `knowledge-platform`, `observability`, and `secrets`. | The `knowledge` example, which is being retired with its project and is excluded from this test set (Section 4.2). |
| Every derivation these examples use, from dependency edges and placement to backups, migrations, network peers, and the path of every generated file. | The text of the generated files, which is Task 3. |
| The flattening's extensions, the final pass, and the `Pinned input set` metamodel and grammars. | Applying files to a running cluster (Task 3 responsibility) or targeting multiple clusters (not supported by current design). |

Table 2: Scope of Task 2

## 2 Alternatives and Justification

### 2.1 Existing Kubernetes Modelling Tools

Three open-source tools were reviewed to see whether they could produce the `Resolved Deployment` from `Project Intent` and `Platform Intent`, or at least take part in the pipeline. Table 3 compares them.

| Tool | What it does | Why it cannot replace the resolution | Where it could help |
|---|---|---|---|
| KubeDiagrams [1] | Draws diagrams from Kubernetes files or a live cluster, mapping resource kinds to icons. | It needs Kubernetes resources as input, so it cannot derive them from deployment intent. | It could visualise the generated files to show links produced by a `Project` document. |
| KubeOrch [2] | A drag-and-drop editor whose back end wires ports, fills templates, writes files, and deploys. | It has no metamodel and no constraints. Its rules are Go code, and it deploys directly, not via Flux. | Its visual graph notation for service connections could inform a `Project` document editor. |
| kubert [3] | A command-line tool to switch cluster contexts and run commands on multiple clusters. | It is an operations tool and does not model or generate anything. | It could apply files to several clusters, which is out of scope. |

Table 3: Existing Kubernetes Modelling Tools

KubeOrch comes closest: it maps service links to addresses and ports. However, it has no shared platform policy or pinned-input model from which to derive placement, locked images, or backups, and no resolved intermediate model to test. KubeDiagrams could inspect the Task 3 output, while a KubeOrch-style editor could author `Project Intent` using the same Ecore metamodel as the transformation.

### 2.2 The Transformation Language

The assignment permits ATL or QVT-Operational (QVTo). QVTo was chosen over ATL because it allows a transformation to read its own output within the same operation. That reading is essential for computing a release identifier: a `Revision` must cover the completed `Application`, including the values derived for its `Process`es, routes, and `ReleaseGate`. ATL's target models are write-only, so this calculation would need to reconstruct the resolved values from the source or run a further transformation over the output. QVTo lets the deployment mapping read the completed target model in its `end` section. This keeps the release identifier with the operation that constructs the deployment and gives it access to the values that operation has produced.

The required construction order also favoured QVTo. Resolved `Process`es must exist before routes and `ReleaseGate` members can refer to them. The `Revision` must be computed after these objects have been populated. Explicit mapping calls make those prerequisites visible in the caller, with trace lookup supplying the references to previously created objects. ATL's called rules could also express an execution order. The reason for preferring QVTo was the combination of that control and access to the populated target, rather than sequencing on its own.

This choice makes correct ordering the transformation author's responsibility. Adding a reference requires its target to be created first, and adding a value covered by the `Revision` requires it to be assigned before hashing. That responsibility was accepted because the dependencies are known from the deployment design and can be reviewed together in the calling mappings.

### 2.3 The Structure of the Transformation

Every derived value must be recorded in a model before code generation. This allows tests to compare the values with the oracles independently of the templates that write them to files.

#### 2.3.1 Flattening Before the Resolution

A single transformation from authored model to `Resolved Deployment` would work, but a flattening precedes the resolution. The flattening copies shared declarations down to each `Process`, including API access and engine `Grant`s (declared access to a secret of a database or transit engine). For example, a `Project` startup budget of 30s reaches every `Process` except one that declares a more specific value, such as 20s. Resolution can then read the selected budget directly from each `Process`, without searching its ancestors. QVT specification Section 8.1.13 likewise composes a preparation transformation before the main one [4]. However, the reason for this split is to isolate declaration precedence from deployment derivation. Its cost is an additional model

and the `EffectiveProject` and `EffectiveApplication` classes, whose copying and override behaviour needs its own tests.

### 2.3.2 Resolving One Project with All Projects as Input

One resolution over all `Projects` at once could produce one `Resolved Deployment` for the whole estate. Instead, each invocation resolves one `Project` into its own model while reading every `Project`'s documents from the same input extent. This preserves a separate output and oracle comparison for each `Project`, while the shared input supplies the cluster-wide view needed for dependencies, routes, and network access. It does not make a `Project` independently resolvable: a change elsewhere can affect its output. Values that depend on all resolved outputs, including the digest over the Vault documents, require the final pass over those models (Section 3.7). A single estate-wide output could compute these values in the same invocation but would combine the output and comparison units.

### 2.3.3 Typed Extents

A rule also has to find the run's other documents. An unqualified `allInstances()` lookup searches all loaded models, which is problematic when both authored (original) and flattened (copied) `Processes` are present simultaneously. Instead, the transformation selects roots by type from the named `intent` and `pinned` extents and stores them in transformation properties such as `projects` and `lock`. Rules navigate from these properties, making their input scope explicit (Section 3.2).

### 2.3.4 Rejecting Unresolvable Inputs

A default for a value the resolution cannot derive would give a file that looks valid and deploys the wrong configuration, so the run aborts instead, through a fatal assertion in an abort query (Section 3.8). The caller then has to handle the failure.

## 3 Solution Design

Task 2 uses three QVT-Operational transformations of varying sizes. The flattening (`lower.qvto`) is 340 lines with 37 mapping operations and 8 queries. The resolution (`resolution.qvto`), the main transformation, is the largest at about 1700 lines (roughly five times the flattening), with 8 mapping operations and 140 queries. The final pass (`estate.qvto`) is the smallest at about 130 lines, with no mappings and 1 helper.

Figure 2 shows the metamodels each transformation reads and writes. The three transformations differ in kind, and all are one-way.

- The flattening is endogenous and horizontal: it reads and writes the source metamodel at the same abstraction level, producing a new model. One `Project` goes in, one comes out.
- The resolution is exogenous and vertical, a refinement: it reads source and pinned-input metamodels and writes the target metamodel, taking several input models and producing one output.
- The final pass is endogenous and in-place: it updates the resolved models of a run through an `inout` extent.

### 3.1 The Flattening

The flattening, `lower.qvto`, maps each authored `Project` onto an `EffectiveProject`. Every `Process` carries all shared declarations that apply to it, while the `Project` and `Application` carry none. Task 2 extended the Task 1 implementation with API access and engine `Grants` (declared access to a secret of a database or transit engine).

Its 37 mappings are nearly all simple copies of declarations from parent to child. Its queries perform the merging logic. A list such as `secrets`, dependency edges, configuration files, or environment entries becomes the union of declarations from `Project`, `Application`, and `Process` levels. If several levels declare the same item, the most specific declaration wins. A single value, such as the startup budget or cutover, also comes from the most specific level that declares it. Cutover determines whether the `Application` keeps serving during an update. Memory and CPU remain only on the `Process`.

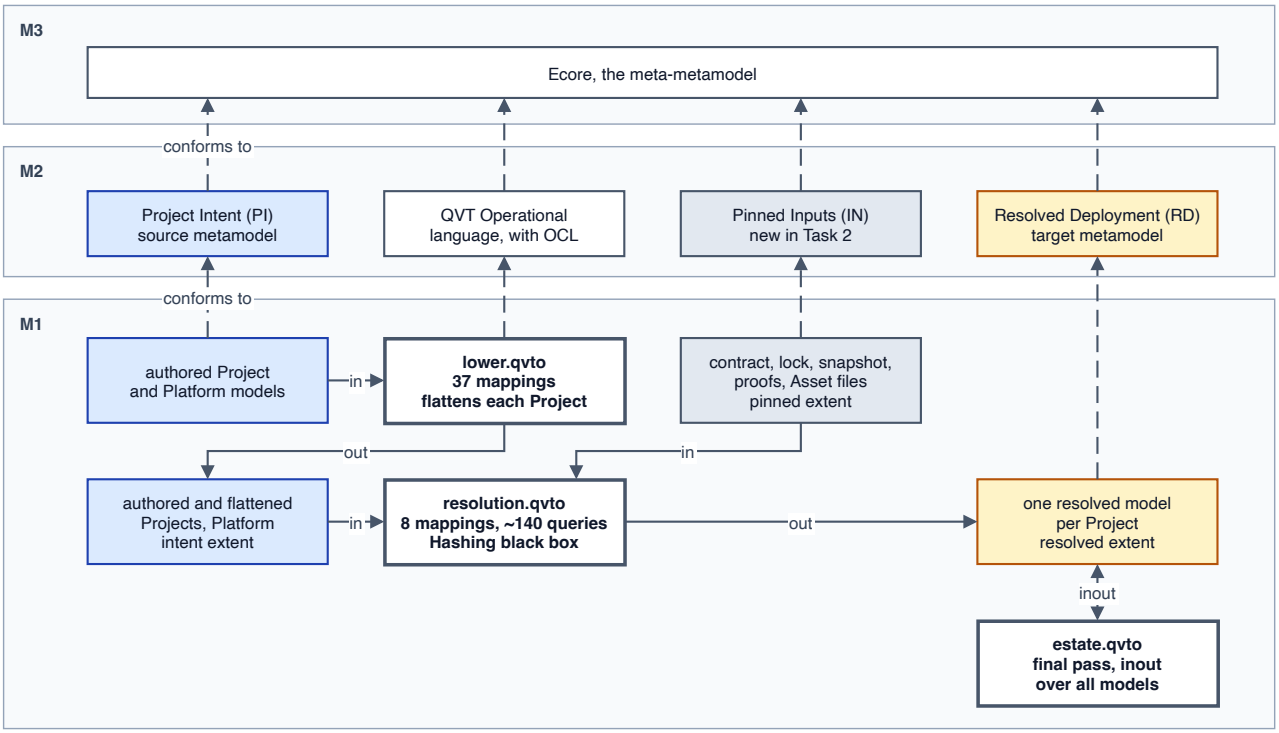



Figure 2: The Three Transformations and Their Metamodels

A shared declaration needs a separate copy in each receiving `Process`. Reusing a cached mapping result would move that object to the last `Process`, because an EMF object can have only one container. Inside `Process::lower`, `self` is the receiving source `Process`. Passing it to each declaration's `copyFor(self)` makes the cache distinguish the required copies. For migrations, a disjuncts mapping dispatches to the copy operation for the source subtype: managed, self-managed, or no migration. The caller can copy any migration without repeating that subtype selection.

The flattening has no aborts of its own: it only copies and merges declarations after input checks. Missing values needed for deployment, such as an unlocked image, are rejected later by the resolution.

### 3.2 Interface, Mappings, and Queries

Example 1 shows how the inputs are grouped: `intent` contains the deployment declarations, `pinned` contains their supporting inputs, and `resolved` receives the target model.

```
transformation resolution(in intent : PI, in pinned : IN, out resolved : RD)
        access library Hashing;
configuration property project : String;     -- the project this run resolves
property projects : Sequence(EffectiveProject) =
        intent.rootObjects()[EffectiveProject]->sortedBy(p | p.project)->asSequence();
property lock : ImagesLock = pinned.rootObjects()[ImagesLock]->any(true);
-- and likewise contract, snapshot, proofs and assetFiles
main() {
        projects->select(p | p.project = project)->any(true).map deployment();
}
```

Example 1: The Signature and Inputs of the Resolution, Abridged

The intent extent holds, for every `Project` in the run, its authored model and its `EffectiveProject`, plus the `Platform Intent`. Authored models are kept only because `Platform Intent` references them via Ecore cross-references. Every rule reads the `EffectiveProject` models.

The pinned extent holds the node contract, images lock, `ClusterState` snapshot, each `Project`'s migration proof (the schema version its migration was tested against), and one `AssetFile` per configuration file. Two con-

figuration properties supply the `Project` name to resolve and the schema-package digest (hash of the metamodel version) to include in the provenance record.

Properties such as `projects` and `lock` select these inputs once. Another selects the `Platform Intent`. A dependency lookup navigates through `projects.applications.processes`, so it searches only the flattened models supplied to this run. Sorting `Project`s by name makes the result independent of input file loading order.

Mappings are used for objects that other rules must reference. For example, the trace connects a source `Process` to its `ResolvedProcess`, allowing a route to reuse that result through `resolveone`. A probe belongs only to its parent and needs no such lookup, so a query creates it with an object expression instead. Each mapped target type has one mapping, making the source and result type sufficient to identify the trace entry. Callers filter elements with `->select` before invoking a mapping. No mapping uses a `when` guard or inherits from another.

Table 4 lists the eight mappings and their purpose.

| **Mapping** | **Creates, and why it is a mapping** |
|---|---|
| `EffectiveProject::deployment` | The `ResolvedDeployment`. Its end section sets values that need the complete model. |
| `EffectiveProject::unit` | The `Project`'s `ReconcileUnit`. Each `Application` refers to it. |
| `EffectiveProject::provenance` | The digest of every input and the hash over them. Its end section computes the hash. |
| `EffectiveApplication::application` | A `ResolvedApplication`. Generated files and the `Revision` refer to it. |
| `EffectiveApplication::gate` | The `ReleaseGate`'s inputs. The `Application` and its configuration refer to them. |
| `EffectiveApplication::migrationOf` | The database migration. Its network policy refers to it. |
| `Process::process` | A `ResolvedProcess`. Gate members, routes, scrapes, policies, and generated files refer to it. |
| `Exposure::exposure` | A `ResolvedExposure`. The generated route file refers to it. |

Table 4: Mapping Operations of the Resolution

Queries calculate values without modifying the input models. They return scalars such as deadlines, new contained objects such as backup plans, or the target spelling of an enumeration value, for example `businessHours` as `business-hours`. The calling mapping assigns the result. Abort queries terminate the run when a value cannot be derived (Section 3.8). The final pass uses a helper instead because it adds files to an existing output model.

Mappings run in a fixed order. The deployment mapping creates the `Project`'s `ReconcileUnit` and local references to prerequisite units first, then `Application`s, provenance, and the path plan (the step that gives every generated file its path). Each `Application` maps its `Process`es before its `Exposure`s, `ReleaseGate`, and migration, because those refer to `Process`es through the trace.

### 3.3 Worked Example

The `minimal` example declares one `Application`, `notes`, with one `Process`, `notes-api`, in about 35 lines of YAML. Table 5 follows declarations through to derived values.

| Declared | Rule | Derived in the resolved model |
|---|---|---|
| `image: notes-api` | `locked` | `ghcr.io/jorisjonkers-dev/notes/notes-api@sha256:9c1e...`, user/group 1000, from images lock |
| `startupBudget: 20s` | `deadlineOf`, startup probe | deadline 60s (three times the budget, per platform policy). The startup probe polls the liveness target every 5s and fails after 4 attempts |
| `cutover: continuous` | `switchoverOf`, `gate` | blue/green `ReleaseGate` with 1 member, deadline 60s, cadence from the `Platform Intent`, error/latency checks |
| `memory: 256Mi, cpu: 50m` | `eligible` | all 7 machines in the node contract, each tested for twice the request size for blue/green deployment |
| `audience: anonymous` | `tierFor`, `exposure` | entry point `public-frankfurt` with TLS certificates issued through ACME, proxy `traefik-public` in the `edge-system` namespace |
| route `/`, prefix | `precedenceOf`, `middlewareOf` | precedence 1, security headers with strict content policy |
| `runtime: node` | `profileOf` | environment variables `OTEL_SERVICE_NAME`, `PORT=8080`, `DEPLOYMENT_ENVIRONMENT`, collector address from `observability` |
| scrape on `http` | `ingressOf`, `monitorOf` | ingress from `traefik-public` and `prometheus` on port 8080, pod monitor for blue/green |
| (nothing) | path plan | 12 generated-file entries (`Deliverable`), including `apps/notes/namespace.yaml` and `estate/edge/public-frankfurt/notes-public.yaml` |

Table 5: Declarations of `minimal`, Followed Through the Resolution

Four values depend on other run documents: the image comes from the images lock, eligible machines come from the node contract, the proxy comes from the `edge Project`, and the collector address comes from the `observability Project`.

Example 2 shows the authored `Process` in YAML, and Example 3 shows the `Process` the resolution assembled from it, in XMI.

```
- name: notes-api
  image: notes-api
  placement: { memory: 256Mi, cpu: 50m }
  startupBudget: 20s
  cutover: continuous
```

Example 2: The Authored `Process` `notes-api`, in YAML (Condensed)

```
<processes name="notes-api" deadline="60s" switchover="blue-green"
    image="ghcr.io/jorisjonkers-dev/notes/notes-api@sha256:9c1e5a4b..."
    uid="1000" gid="1000" memory="256Mi" cpu="50m">
  <startup period="5s" timeout="5s" failures="4"> ... </startup>
  <placement> <eligibleNodes>enschede-t1000-1</eligibleNodes> ... </placement>
</processes>
```

Example 3: The Resolved `Process` `notes-api`, in XMI (Abridged)

The path plan assigns the `Canary` resource, used by the Flagger rollout controller to manage blue/green replacement, to `apps/notes/notes/canary.yaml`. Per-`Application` files follow `apps/<project>/<application>/`. Here both names are `notes`. Section 4.1 explains the complete-model comparison.

### 3.4 The Derivation Rules

Table 6 lists the rules by target element. Each implements a derivation from the specification.

| Target value | Rule | Derivation |
|---|---|---|
| *Project* | | |
| reconcile unit and order | `unit`, `afterOf` | Create the `Project`'s unit. Its dependency list tells Flux to wait for provider `Project`s' units and, where needed, the secrets-provisioning unit. |
| *Application* | | |
| Secret Store (Vault) | `synced`, `storeEndpoint` | Record the address consulted when a secret, backup, or migration must be resolved. |
| migration | `migrationOf`, `datastoreOf` | Combine the locked runner image, schema version from the migration proof, and database address from dependencies. Derive its credential and permitted connections. |
| release gate | `gate`, `switchesBlueGreen` | Register each blue/green `Process` with readiness as a gate member. Use the largest member deadline and platform analysis settings. Enable error-rate and latency checks for members exporting telemetry. |
| *Process* | | |
| image, user, group | `locked` | Select the lock entry for the image name. |
| switchover | `switchoverOf` | Use stop-start when interruption permits it, rolling for the `Application`s the platform itself delivers, and blue/green otherwise. |
| deadline | `deadlineOf` | For a preparation job (lifecycle `prepare`), round the startup budget up to whole seconds without multiplying it. For other lifecycles, multiply by three and round down. Without a budget, use 600s. |
| probes | `Probe::cadenced` | Combine the declared target with the period, timeout, and failures from the `Platform Intent`. The startup probe polls the liveness target every 5s until the budget runs out. |
| eligible machines | `eligible` | Keep nodes meeting the declared placement requirements and CPU/memory requests, including sidecars. Continuous replacement requires capacity for both old and new copies. |
| environment | `environmentOf`, `entryFor` | Cluster-specific entries replace base entries of the same name. Resolve placeholders, add runtime-profile variables, and sort by variable name. |
| secrets | `Grant::resolvedFor` | A key-value `Grant` retains its path and keys, and names the Kubernetes Secret to synchronise. A transit or database `Grant` records the Vault paths its policy permits. |
| volumes and backups | `backupFor`, `backupEgress` | For volumes requiring backups, select the platform schedule, retention, and engine-specific image. Derive a separate service account, claim, database connection, credential, and off-cluster destinations. |
| configuration files | `Asset::resolvedFor` | Record the file text, named after the `Process`, the file, and a digest prefix, so a changed file gets a new name. |
| dependency edges | `edgeOf` | Select a `Process` providing the named `Application` and port. If none exists, try the platform's external providers. Record its address and permitted network peer. |
| *Exposure* | | |
| entry point | `tierFor` | Select the first platform entry point whose allowed audiences include the `Exposure`'s declared audience. |

| Target value | Rule | Derivation |
|---|---|---|
| route precedence | `rankOf`, `precedenceOf` | Exact routes come before prefix routes, and longer prefixes before shorter ones. |
| middleware | `middlewareOf` | Attach authentication for authenticated routes, security headers, and redirect if declared. |
| *Network policy, per Process* | | |
| ingress | `ingressOf` | Admit route proxies, the metrics collector for a scraped `Process`, dependency consumers, its backup identity, and database migrations. For the Reporter (receiver of rollout events), also admit Flux's notification controller. |
| egress | `egressOf` and resolved connections | Add cluster DNS and, for secret holders, the Secret Store. Dependency edges supply provider peers. API rules supply server ranges. Migration and backup plans carry their own peers and off-cluster ranges. |
| *Deployment* | | |
| provenance | `provenance` | Record each input's digest, then hash that record to identify the complete input set. |
| revision | end of deployment | Hash each `Application`'s complete resolved configuration in the specification's JSON representation (Section 3.6). |
| generated files | `projectFiles`, `applicationFiles`, `edgeFiles`, `vaultFiles` | Record the path of every file and the elements it is generated from (Section 3.4.5). |

Table 6: Derivation Rules, by Target Element

### 3.4.1 The Process Mapping

Example 4 shows how the `Process` mapping combines these rules. The image lock entry supplies the repository and digest, `deadlineOf` supplies the deadline, and `eligible` filters the nodes. Optional fields are assigned only where they apply. For example, a one-off job has no switchover: assigning the enumeration's default value would incorrectly record a blue/green deployment strategy for that job.

```
mapping PI::Process::process(in app : EffectiveApplication, in prj : EffectiveProject)
                : ResolvedProcess {
        init { var own : LockedImage := locked(self.image); }
        name := self.name;
        image := own.repository + '@' + own.digest;
        if self.lifecycle = Lifecycle::application
                then switchover := switchoverOf(app, self) endif;
        deadline := inSeconds(deadlineOf(self));
        secrets := self.secrets->collect(g | g.resolvedFor(self));
        placement := object ResolvedPlacement {
                eligibleNodes := contract.nodes->select(n | eligible(self, n))->collect(n
    | n.name);
        };
        dependencies := self.dependsOn->collect(e | edgeOf(e));
        environment := environmentOf(self, prj)->sortedBy(e | e.name);
}
```

Example 4: The Process Mapping, Abridged

### 3.4.2 The Edge Rule

A dependency edge names an `Application` and a `Surface` (named port). For example, an edge to `platform-postgres` names its `postgres` port. The query `edgeOf` finds the first `Process` in the cluster

belonging to that Application that provides that Surface. The resolved edge records the Process address and the network peer the consumer may reach. If no Process provides the Surface, the Platform Intent external provider list is tried as a fallback. This mechanism allows the auth Project to reach a mail server outside the cluster. If neither has it, the run aborts.

During migration to this deployment system, the HandoverLedger in the Platform Intent records where Applications still served by the previous system run. Their resolved addresses and peers use the recorded namespace and workload name until they move.

### 3.4.3 Environment Variables and Placeholders

A Process's environment files consist of a base file and, optionally, one overlay per cluster. The resolution selects the overlay named by the Platform Intent's cluster, then replaces base entries with overlay entries of the same name. Other base entries remain.

An entry may contain a placeholder, such as DB_HOST=${dependency:platform-postgres.host}. The query Placeholder::entryFor handles four kinds. A secret placeholder becomes a reference to a key of a secret the Process holds, so the value never appears in a file. A dependency placeholder takes the host or port of the Process dependency edge from edgeOf. An exposure placeholder takes the URL, host, or scheme of an Exposure anywhere in the cluster. An identity placeholder takes the Process name, namespace, or Vault role (the binding that grants its service account a secret-access policy). Text after a placeholder is kept, as in AUTH_LOGIN_URL=${exposure:auth.public#url}/login.

The runtime profile then adds the platform-defined variables for the selected language runtime, such as PORT. Declaring one of these names in an environment file aborts the run, because it would conflict with the platform-defined value.

### 3.4.4 Network Peers

The ingress policy of a Process combines peers from six sources (Table 6). Most require looking across Projects: consumers with a dependency edge to the Process, migrations that access its database, proxies, and the metrics collector named by the Platform Intent. Each source yields tuples of rule, namespace, Process, and port. Distinct PolicyPeer objects are not considered equal even if their field values match, because they are separate model elements. The union therefore uses tuples and asOrderedSet() to remove duplicates before creating PolicyPeer objects. The rule name remains part of each tuple, so two rules allowing the same connection retain both rule identifiers.

### 3.4.5 The Path Plan

The path plan represents each generated file as an instance of a Deliverable subclass, recording its path and the model elements used to generate it. Each Project gets a namespace file, index, and default-deny network policy. If an Application synchronises secrets, a ConnectionFile (for the VaultConnection resource) tells the secret operator (the component that syncs secrets into the cluster) which Vault address to read.

Each Application gets a directory with one file per resource kind:

- a workload file;
- a service account;
- RBAC;
- volume claims;
- a configuration map carrying ReleaseGate inputs as JSON;
- backups;
- a migration;
- a monitor;
- synced secrets;
- a network-policy file with one policy per Process.

Blue/green Processes add a Canary file. Others add a Service file. Multi-replica Processes add a disruption budget. Empty files are not planned, as they would generate unnecessary Kubernetes resources.

Each index lists the files Flux should apply from its directory. Network policies are included in the generated output for review, but excluded from these indexes while the specification's policy stage is render-only. This allows their proposed connections to be inspected before enforcement. Each `Exposure` gets a route file under its entry point (Traefik listener). Each service account holding a secret gets a Vault policy and role.

Paths can therefore be inspected and compared before rendering. The transformation has no separate path-collision abort. Templates receive these paths rather than choosing them.

### 3.5 The Trace and End Sections

Gate members, routes, metrics scrapes (monitoring configurations), and per-`Process` files all refer to a `ResolvedProcess`. They find it by calling `p.resolveone(ResolvedProcess)` on the source `Process`, which the `Application` mapping has already transformed (Section 3.2).

The `end` sections set values whose prerequisites are created by the mapping body. The `Revision` must wait until the `Application` is complete. The migration job tag then takes a prefix of that digest so both identify the same release. Similarly, the render hash is computed after all input digests have been recorded, and `ReconcileUnit` dependencies are assigned after their local reference objects exist. Each output contains its own unit and name-only `ReconcileUnit` objects for prerequisite units in other `Project`s. The `after` reference points to those local objects, not into another output model.

### 3.6 The Black-Box Library

A digest is a SHA-256 hash over canonical text. OCL lacks a byte-hashing operation, so the Java class `Hashing` provides three operations that can be called from OCL expressions. `digestOf` walks any model element reflectively, writes canonical JSON and hashes it for input provenance. `textDigestOf` hashes a file's text, for the content-derived names of configuration files. `rawDigestOf` hashes a string as-is.

The `Revision` uses `rawDigestOf` to hash the canonical JSON representation directly. Generated files carry this digest, so it must match the oracle exactly. The specification hashes the `Application` configuration in canonical JSON (RFC 8785). That specified JSON has a different structure from the Ecore target model. The OCL query `elementJson` therefore constructs the required JSON from the source and target models, sorting keys, omitting absent values, and using the specified representation of each value. The black box only hashes this text. The transformation determines which values it contains.

Calls explicitly cast model elements to `EObject` before invoking the library's contextual operations. The standalone runner and Eclipse plug-in both register the `Hashing` library, and the QVTo modules access it by import.

### 3.7 The Final Pass

One job installs the Vault policies and roles from every `Project` into the Secret Store. Its name contains a digest of all these documents. Changing a document therefore gives the next deployment a new job name, causing the updated configuration to be installed.

The final pass, `estate(inout rendered : RD)`, runs once over all `Resolved Deployment`s produced together. It collects every `VaultPolicyFile` and `VaultRoleFile`, writes each as canonical JSON in OCL, sorts them by file name, hashes the whole with `rawDigestOf`, and plans the job, cronjob, configuration map, service account, and network policy. These files and their index are stored in the first model in the caller's project list, so the job is generated once for the combined output.

The pass also plans one cluster-wide route forwarding Flux events to the Reporter. All resolutions normally share one `Platform Intent`. A defensive assertion rejects different job namespaces or images if the pass is called directly with models from different runs. It does not compare every job setting.

### 3.8 Aborts and Running the Transformations

Fifteen fatal assertions in the resolution abort the run when a value cannot be derived. Thirteen are in abort queries that a rule calls instead of returning an invalid value. For example, `locked` returns the lock entry for an image or calls `unlocked`, whose assert fatal ends the run with `E_UNLOCKED_IMAGE`. Two check `ReleaseGate` preconditions at the start of the gate mapping. The final pass has one more, for the inconsistent job namespaces or images described above.

The checks reject missing inputs and providers, unsupported quantities, unresolved placeholders, invalid secret references, and author-defined runtime-profile variables. They also reject backups without schedules or engines,

migrations without a database, blue/green deployments without readiness or delivery services, and `Grant`s without a Secret Store.

Where the specification defines an error code, the message contains it. An audience unsupported by any entry point needs no abort here: the `E_NO_TIER_FOR_AUDIENCE` constraint already rejects it before the transformation runs.

Three Java classes (`Lowering`, `Resolution`, and `Estate`) run the transformations through QVTo's own `TransformationExecutor`. They register the black box, set configuration properties, pass the extents, and return the result or fail with the executor's diagnostic. `Lowering` is the implementation name of the flattening runner. The flattening does not change the abstraction level.

The build executes the transformations with QVTo's standalone executor on every code change. The archive's `README.md` gives terminal and Eclipse instructions for all three passes. It also maps each example to its output project and explains the generated model extents.

## 4 Tests and Evaluation

The transformations are tested by running them on eight examples and comparing their output with oracle files, which are the expected outputs committed with the specification's worked examples. Agreement provides evidence of conformance to expected values. However, the strength of this evidence varies depending on how each oracle was derived.

The build runs six types of tests on every change:

- **Flattened models.** `auth`, `data`, `knowledge-platform`, and `minimal` are compared byte-for-byte with `expected/effective.json` beside each specification example. These committed fixtures are not regenerated during the tests.
- **Dependency edges.** For each example, the resolved edges are exported as JSON and compared byte-for-byte with the expected file. These tests read only files the pipeline wrote, so they check the pipeline as a user would run it.
- **Revisions.** Each expected resolved file records its `Revision`, a digest of the `Application` element in the specification's canonical JSON form. A matching `Revision` checks the represented decisions, including each `Process`'s ingress and egress peers (Section 4.1.1).
- **Complete expected models.** The adapted Task 1 model of `minimal` and a reviewed regression snapshot of `secrets` are compared using EMF's structural equality (`EcoreUtil.equals`). No attributes or references are excluded. Both expected models record the transformation's provenance digests. Matching those fields checks stability, not agreement with the specification's differently represented inputs.
- **Generated files.** Tests run the Task 3 templates on the transformation output and compare every generated file with its committed counterpart. This checks the resolved values used by the templates as well as their serialisation.
- **Interface tests.** JUnit tests check that the resolution accepts a minimal `Project`, rejects a `Project` absent from the input, and reports the file name when a configuration file is missing. Final-pass tests check that one Vault policy job covers two `Project`s, its name changes when a document changes, no job is generated without documents, and conflicting job definitions abort the run. Eight flattening cases check the merging of shared declarations.

Coverage and mutation tools require full line and branch coverage and a 100% mutation score for the Java integration code. They do not measure QVTo, so the comparisons establish no QVTo branch-coverage result.

### 4.1 Results

Table 7 lists the eight examples. All resolve successfully and every comparison passes, including the complete-model comparison of `minimal` and `secrets`. The full build passes 323 tests. Its oracle suite contains 81: 10 parsed-intent comparisons, 4 flattened models, 47 input refusals, 8 composition refusals, 8 dependency lists, 1 rendered tree, 1 metamodel descriptor, and 2 checks on the test setup. This total includes Task 1 parsing and constraint tests.

"Empty" means the expected dependency lists contain no edges. Only the `auth`, `delivery`, and `knowledge-platform` examples compare non-empty edge sets. Revision fractions are matching digests over the total number of `Application`s. For `knowledge-platform`, three of its four `Application`s lack committed oracle digests, so

| | Counted in the resolved model | | | | | | | Equal to the expected output | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Example** | **Apps** | **Proc.** | **Edges** | **Secr.** | **Vol.** | **Env.** | **Files** | **Edges** | **Revisions** | **Files** | **Model** |
| `minimal` | 1 | 1 | 0 | 0 | 0 | 6 | 12 | empty | 1/1 | yes | yes |
| `auth` | 1 | 2 | 4 | 3 | 0 | 21 | 19 | yes | 1/1 | yes | – |
| `data` | 3 | 3 | 0 | 1 | 3 | 3 | 37 | empty | 3/3 | yes | – |
| `delivery` | 4 | 4 | 1 | 0 | 0 | 0 | 27 | yes | 4/4 | yes | – |
| `edge` | 2 | 2 | 0 | 0 | 0 | 0 | 13 | empty | – | – | – |
| `knowledge-platform` | 4 | 7 | 5 | 4 | 6 | 84 | 49 | yes | 1/4 | yes | – |
| `observability` | 2 | 2 | 0 | 0 | 0 | 0 | 15 | empty | – | – | – |
| `secrets` | 1 | 1 | 0 | 0 | 0 | 0 | 8 | empty | – | – | yes |

Table 7: The Resolved Examples and Their Comparisons (–: no committed oracle of that kind)

only one is compared.

Taken together, the examples exercise the derivations in Table 6. `minimal` exercises the common `Process` and `Exposure` rules (Table 5). The migration, placeholder, and external-provider rules run in `auth`, which holds three secrets and reaches a mail server outside the cluster. The volume, backup, and configuration-file rules run in `data`, which holds one volume of each durability class (reconstructible, recoverable, irreplaceable), a `Sidecar`, and a configuration file. The rolling and stop-start switchover rules run in `delivery`, which also has API access. `secrets` models the Secret Store, `knowledge-platform` is the largest example, and `edge` and `observability` supply the proxies and collectors that other examples look up.

#### 4.1.1 Revision Evidence

An `Application`'s canonical JSON includes its `Process`es (with images, identities, probes, placement, secrets, volumes, backups, environment, dependencies, and network peers), its `Exposure`s and routes, its `ReleaseGate`, and its migration. Provenance, file paths, typed resource objects, and the estate-wide policy job fall outside it.

The transformation builds that text from its own `Resolved Deployment` (Section 3.6), so one different value, one extra environment variable, or one different order changes the digest. Ten `Application`s across five examples have a `Revision` in an expected file, and all ten match.

The comparison checks this representation as a whole, without locating a mismatch or checking omitted values. A further test checks sensitivity: increasing memory in `minimal` from 256Mi to 512Mi changes `Revision`, whereas changing only the capture time of the `ClusterState` snapshot leaves it unchanged.

#### 4.1.2 Generated-File Evidence

Generated-file comparisons additionally check typed resource serialisation, Vault documents, and destination paths. Although `Revision` covers a `Process`'s network peers, it does not check their serialisation as Kubernetes policies. Across five examples and the cluster-wide output, all 167 committed files are reproduced byte-for-byte, including the final-pass job files.

#### 4.1.3 Gaps in the Evidence

Two examples, `edge` and `observability`, have no `Revision` oracle, no committed generated files, and no expected model. Their `dependencies.json` files list no dependency edges, meaning these `Project`s have no dependencies on other `Application`s within the cluster. Evidence for `edge` and `observability` is therefore limited to successful execution and to checking that examples using their proxies and collectors produce correct results. Those check the values consumed by other `Project`s, leaving the remaining configuration unchecked.

The `secrets` snapshot has no independent JSON counterpart or committed generated files. Reviewing the transformation's output to obtain that snapshot found a specification gap: the Secret Store network policy allows no inbound traffic, although holders of secrets are allowed to send traffic to it. If enforced, the receiving policy would still block those connections. Permission on the sending side is insufficient.

The Task 1 model of `minimal` was adapted to the metamodel changes in Appendix A.2. Its `Revision` was copied from the oracle, so that field and its recorded provenance provide regression checks rather than independent

derivations.

#### 4.1.4 Defects Found

One directory-layout defect required a correction: cluster-wide route files were planned under `apps/edge/`, inside the `edge` `Project`'s directory. This mixed ownership of a directory without requiring two files to claim an identical path, so duplicate-path checks could not detect it. Cluster-wide files moved to `estate/`. The specification's path rules, path plan, and expected files changed together. Agreement after that correction guards against regression. It is not an independent test of the revised path rules, which required reviewing who consumes each directory.

### 4.2 Limitations

Agreement with the oracles supports the transformation's conformance to their expected outputs. It does not establish that the oracles or the specification are correct. The hand-written Task 1 model of `minimal` supplies a partly independent check. Its adapted fields have the qualifications above.

The examples exercise the derivation families, but a matching result for one input does not prove that every branch works. The `knowledge` example was excluded because that `Project` is being retired and its successor `knowledge-platform` was selected instead. The suite does not attempt its resolution, so exclusion supplies no evidence that it succeeds or that retirement is unsupported.

During Task 2, checks for secret references, placeholders, and platform requirements were added to the constraints in `project-intent.ocl`, before resolution. These errors had previously reached QVTo aborts. Remaining aborts stop at the first problem without a source location. Cross-document constraints still use `allInstances()`. Navigation from a single root object remains planned.

Placement produces eligible nodes for each `Process`. Kubernetes chooses the actual node. Each eligibility check considers that `Process` in isolation, so several may be eligible for a node even though their combined requests exceed its capacity.

Ordering between `ReconcileUnit`s is defined at `Project` granularity because Flux orders these groups of files as whole units. Individual `Application`s within a unit cannot be ordered separately.

Provenance uses the EMF representation of inputs, so its digests differ from the specification's. Whole-model comparisons pass because the expected EMF models record those computed values, not because provenance is omitted. Digests used in generated files do match: `Revision`, configuration-file names, and the Vault policy job name.

Automated tests use the standalone executor. The Eclipse launch configuration and `plugin.xml` registration have only been checked manually.

### 4.3 Reflection

Flattening made resolution rules simpler: each reads one selected value from a `Process`, without repeating the precedence rules for shared declarations. It also introduced an intermediate model and copying rules that need their own tests. A failed merge comparison can be investigated without involving image or provider resolution. This separation makes the extra model worthwhile.

Resolving one `Project` at a time kept its result inspectable, but did not remove the need to reason about the estate. Cross-project references still require the full input extent, and a digest over all Vault documents requires the final pass. The split therefore helped local comparison at the cost of another transformation and a rule for where shared output is stored.

QVTo's explicit sequencing suited creation of `Process`es before their routes and computation of `Revision` last. Its trace avoided a manual source-to-target table. ATL also offers traces and explicitly invoked rules, so these results support the chosen organisation without establishing that ATL could not express it. No equivalent ATL implementation was built.

The greatest effort went into reproducing `Revision`. The target model has a different structure from the specification's JSON, so the transformation had to reconstruct the specified field names, omissions, and ordering before hashing. That reconstruction occupies about one sixth of `resolution.qvto`. A hash mismatch could mean either a wrong deployment decision or a wrong JSON representation, with no indication of where they differed. A better test design would compare that JSON directly before checking its digest, separating derivation

errors from serialisation errors. Using a Java hash function saved little of this work: most of the code was needed to construct the JSON in OCL.

Evidence should have been planned alongside the examples. `secrets` and `knowledge-platform` ran before their output comparisons were added. Those runs established execution, not correctness of the derived values. Constructing expected results before running the transformation, and retaining the original Task 1 model beside its adapted version, would make independent checks easier to distinguish from regression snapshots.

## References


[1] P. Merle and F. Petrillo. Visualizing Cloud-native Applications with KubeDiagrams. In *Proc. VISSOFT 2025*, pp. 106–116, 2025. Tool: https://github.com/philippemerle/KubeDiagrams.

[2] KubeOrch, visual Kubernetes workflow orchestrator. https://github.com/KubeOrch (repositories `core` and `ui`), visited 15 October 2026.

[3] kubert, a kubectx and kubens alternative. https://github.com/idebeijer/kubert, visited 15 October 2026.

[4] Object Management Group. *Meta Object Facility Query/View/Transformation Specification*, version 1.3, 2016.

# A Changes to the Task 1 Metamodels

This appendix lists the changes between the Task 1 metamodels and those the Task 2 transformations read and write. The comparison checked each class and feature in the `.ecore` files. The diagrams are grouped at the end of this appendix.

A third metamodel for the `Pinned input set` (read-only documents not written by `Project` authors) was added (Appendix A.1). The target metamodel gained derived values and twelve kinds of generated file (Appendix A.2), and the source metamodel gained declarations that the other examples need (Appendix A.3). Four grammars were added (Appendix A.4).

No class was removed, but some features were removed or replaced, including document headers on `EffectiveProject` and single-`Process` references on generated-file classes. The Task 1 hand-in showed OCL helper definitions such as `levelsAbove` and `derivesDatabase` as derived attributes in the diagram. They remain OCL definitions in `project-intent.ocl` and were never Ecore features.

## A.1 The Pinned Inputs Metamodel

The `pinnedinputs` package models the additional documents read by the resolution (Section 1.3).

The node contract (the list of machines and their capacity) is generated from the platform team's machine descriptions. Other programs publish the images lock (each image pinned to a digest) and the `ClusterState` snapshot. A `Project` build writes its migration proof (the schema version its migration was tested against).

A stored JSON description of the source metamodel's classes and features is compared with the specification's authored-document schemas. Pinned inputs are kept in a separate metamodel because they are not part of those schemas. The target metamodel has no references to pinned-input classes: the `Resolved Deployment` records the values derived from these inputs and their digests instead.

Figure 3 shows the metamodel. Table 8 shows which rules read each class.

| Class | Read by |
|---|---|
| `NodeContract`, `Node`, `Allocatable`, `Gpu`, `Disk` | `eligible` compares resource requests with available capacity and checks hardware requirements. CPU quantities such as 53500m denote 53.5 cores. |
| `ImagesLock`, `LockedImage` | `locked`: every image of a `Process`, `Sidecar`, backup method, migration runner, and Vault policy job. |
| `ClusterState`, `Binding` | `boundOf`: the machine that already holds a `Volume`. |
| `CurrentPlacement`, `RollbackBackup` | Retained as snapshot data for delivery operations. These classes do not affect the resolution's output. |
| `MigrationProof`, `ProvenApplication` | `migrationOf` records the tested schema version. `gate` also reads whether the release contains non-transactional changes. |
| `AssetFile` | `Asset::resolvedFor` reads the configuration-file text. Each file is wrapped in an `AssetFile`, rather than parsed with an Xtext grammar. |

Table 8: The Classes of the Pinned Inputs Metamodel and the Rules That Read Them

Disk medium (field `media`) and architecture are strings in the node contract, although the source metamodel restricts `Process` requests to enumerations. The contract can describe hardware, such as an sdcard, that no `Process` may request. Reusing the source enumeration would prevent such nodes from being represented. Eligibility therefore compares each requested enumeration's spelling with the corresponding contract string.

## A.2 Changes to the Target Metamodel

Both halves of the target metamodel grew: the values the resolution derives for each `Application`, and the generated-file entries of the path plan. Derivation rules read policies from the `Platform Intent`. The target metamodel records the resulting values.

The root class is unchanged, but there is now one instance per `Project` instead of one for the whole cluster (Section 1.3).

**Derived Values.** Figures 4 and 5 show the new or changed derived-value classes, at deployment and application level and at process level. Table 9 lists their feature changes and reasons.

Task 1 described many values in prose but modelled only those needed by its worked example. The other examples also need migrations, backups with database connections, secrets synchronised into the cluster, API access, `Sidecar`s, configuration files, and two cluster-wide values: the Vault policy job, which writes every Vault policy, and the route of Flux's events.

| Class | Change | Reason |
|---|---|---|
| `ResolvedDeployment` | added `policyJob`, `events` | The Vault policy job and the route for Flux's events belong to the whole cluster. Every `Project` model records the same job. The final pass plans their files. |
| `ResolvedPolicyJob`, `ResolvedEvents` | new | The policy job records its identity, image, role, resources, and egress. The events route records the Reporter's namespace and address. |
| `ResolvedApplication` | added `secretStore`, `scrape` | Record the Vault address for secret synchronisation and the metrics scrape settings. Both contribute to the `Revision`. |
| `ResolvedScrape` | new | The scraped `Process`, `Surface` (named port), path, interval, and timeout. |
| `ReleaseGate` | added `endpoint`, `migration` | An address that every `Canary` (Flagger) webhook calls. The migration the gate starts and may undo. |
| `GateAnalysis` | added `errorRate`, `latency` | The thresholds of the two analysis checks, from the `Platform Intent`. |
| `GateMigration` | new | The migration's service account and what its proof records. |
| `ResolvedMigration` | added `uid`, `gid`, `identity`, `deadline`, `memory`, `cpu`, `scratch`, `database`, `credential`, `egress` | Task 1 named the service account, owner role, and deadline in prose. A migration job also needs the database address, the owner credential that only it holds, and a network policy. |
| `MigratedDatabase` | new | The database host, port, and name. |
| `ResolvedProcess` | added `lifecycle`, `api`, `assets`, `sidecars` | Record whether the `Process` serves continuously, runs once, or prepares a release, plus its API access, configuration files, and `Sidecar`s. These values contribute to the `Revision`. |
| `ResolvedApiAccess`, `ResolvedApiRule`, `AdmittedRange` | new | The API rules a `Process` holds. The address range a policy admits includes the API server for API access and the backup off-cluster destination. |
| `ResolvedAsset`, `ResolvedSidecar` | new | A configuration file with a content-derived name. A `Sidecar` has a locked image and resources. |
| `ResolvedGrant` | `path` and `access` optional. Added `engine`, `paths`, `destination`, `mountAt`, `fileMode`. A transit or database `Grant` covers several Vault paths. A synchronised key-value `Grant` also records the destination Kubernetes Secret and its mount settings. | |
| `PolicyPath` | new | One path of a Vault policy and the operations allowed on it. |
| `ResolvedVolume` | added `mountAt` | The workload template needs the mount path. |

| Class | Change | Reason |
|---|---|---|
| BackupPlan | added uid, gid, identity, claim, egress, peer, credential, destinations | A backup runs as its own service account with its own claim and network policy. It connects to the Process it backs up and copies off-cluster with a credential. |
| BackupPeer | new | Where a backup connects, and the credential it logs in with. |
| EnvEntry | value optional. Added secret. A secret placeholder becomes a reference, so the value never appears in a file. | |
| SecretReference | new | The secret path and key a variable reads. |
| StartupProbe | added timeout | The specification gives the startup probe the Platform Intent probe timeout. |
| ResolvedExposure | added listener, certificates, proxyApplication, proxyNamespace | The entry point's protocol and certificate source. The proxy whose middleware the routes name. |
| PolicyPeer | added rule. process is optional. A peer identifies a permitted network connection. Recording its rule preserves distinct reasons for allowing the same connection. Cluster DNS needs a namespace but no Process. | |

Table 9: Changes to the Derived-Value Classes of the Target Metamodel

**Generated Files.** Figure 6 shows every subclass of Deliverable, the generated-file entry, up from Task 1's 9 to 21.

Workload, Canary, and service account files, which Task 1 tied to one Process, now list all of an Application's Processes, because the specification puts one file per Application holding all its Process resources. For the same reason, the network policy file holds one ProcessPolicy per Process, and the Secret file refers to its Application and holds one SecretHolder per service account, since a backup's service account holds Secrets too. Task 1's worked example had one Process, so this difference did not show.

MonitorFile gained timeout and kind. A stop-start Process is scraped through its Service. Other strategies use a Pod monitor.

The twelve new file kinds are Services, volume claims, configuration maps, backups, disruption budgets, migrations, RBAC, the ConnectionFile (VaultConnection) used by the secret operator, Vault policies, Vault roles, the Vault policy job, and the events route.

The hand-written model of minimal from Task 1 was updated with these changes. Its files refer to the Process through processes, it gained the configuration map that carries the ReleaseGate inputs, and its expected values were adapted as discussed in Section 4.1.3.

### A.3 Changes to the Source Metamodel

The source metamodel kept every class. Table 10 lists what was added or changed.

Most additions make platform policies explicit that Task 1 left implicit. Declaring these values in the Platform Intent keeps them out of the transformation rules, where they would otherwise be hard-coded.

| Class | Change | Reason |
|---|---|---|
| Process | added api (ApiAccess, ApiRule, ApiVerb) | The Process declares the Kubernetes API access it needs, with a reason. The platform's own delivery Applications need it. |

| Class | Change | Reason |
|---|---|---|
| `Platform` | added `apiAccess` (`ApiAccessPolicy`) | Which `Applications` may hold API access, and the API server address range. |
| `Platform` | added `secretStore`, `telemetry` (`TelemetryPolicy`), `policyJob` (`VaultPolicyJob`) | References identify Vault and the telemetry services in other `Projects`. The policy job declares its image, role, and resource requests. |
| `DeliveryPolicy` | added `gate`, `reporter` | References to the `ReleaseGate` service and Reporter `Applications`. |
| `AnalysisPolicy` | added `errorRate`, `latency` | The thresholds the `ReleaseGate` service checks. |
| `Substrate` | added `clusterDns` | The namespace every egress policy admits for DNS. |
| `EnginePolicy` | added `surface` | The port a database engine backup connects to. |
| `OffClusterCopy` | added `egress` (`DestinationRange`) | The address ranges a backup copy goes to. |
| `HandoverLedger` | added `serving` (`ServingLocation`) | Where a `Project` not yet moved still runs, so an edge to it resolves. |
| `RenderedArtifacts` | added `pullSecret` | Used by delivery, not by the resolution. |
| `Placeholder` | added `suffix` | Text after a placeholder, as in `${exposure:auth.public#url}/login`. |
| `Grant` | `rotation` required | Every `Grant` states how it tolerates rotation, which decides its restart targets. |
| `Volume` | `size` required | A claim cannot be generated without a size. |
| `EffectiveProject` | removed `apiVersion`, `kind`, `schemaVersion` | The flattened model is written by the flattening, not by an author, so it has no document header. |

Table 10: Changes to the Source Metamodel

### A.4 Grammars for the Pinned Inputs

The pinned inputs use grammars rather than a Java YAML reader or hand-written XMI, so the transformation can read the examples' YAML directly. No input value changed; one quoting adjustment was needed, as described below.

Four grammars were added beside the `Project` and `Platform` grammars: `NodeContract`, `ImagesLock`, `ClusterState`, and `MigrationProof`. Each inherits the `Project` grammar, reusing its terminals, indentation tokens, and scalar rule, and adds only the rules for its own document. As with the `Platform Intent`, the file name selects the grammar: `node-contract.yml`, `images.lock.yml`, `cluster-state.yml`, and `migration-proof.yml`.

In `cluster-state.yml`, the `Project` grammar scalar terminal cannot contain a colon, so the snapshot's capture time is now quoted, as digests and URLs already were. The parsed string value is identical.

Example 5 shows the smallest grammar.

```
grammar dev.jorisjonkers.deploykit.emf.syntax.ClusterState
        with dev.jorisjonkers.deploykit.emf.syntax.ProjectIntent
import "https://jorisjonkers.dev/deploy-kit/pinned-inputs/1"

ClusterState returns ClusterState:
        (('apiVersion' ':' apiVersion=Text)
        & ('kind' ':' kind=Text)
        & ('schemaVersion' ':' schemaVersion=Text)
        & ('cluster' ':' cluster=Text)
        & ('capturedAt' ':' capturedAt=Text)
        & ('bindings' ':' BEGIN (DASH BEGIN bindings+=Binding END)* END)
```

```
        & ('placements' ':' BEGIN (DASH BEGIN placements+=CurrentPlacement END)* END)
        & ('backups' ':' BEGIN (DASH BEGIN backups+=RollbackBackup END)* END)?);

Binding returns Binding:
        (('claim' ':' claim=Text)
        & ('node' ':' node=Text));
```

Example 5: The Grammar of the `ClusterState` Snapshot, Abridged

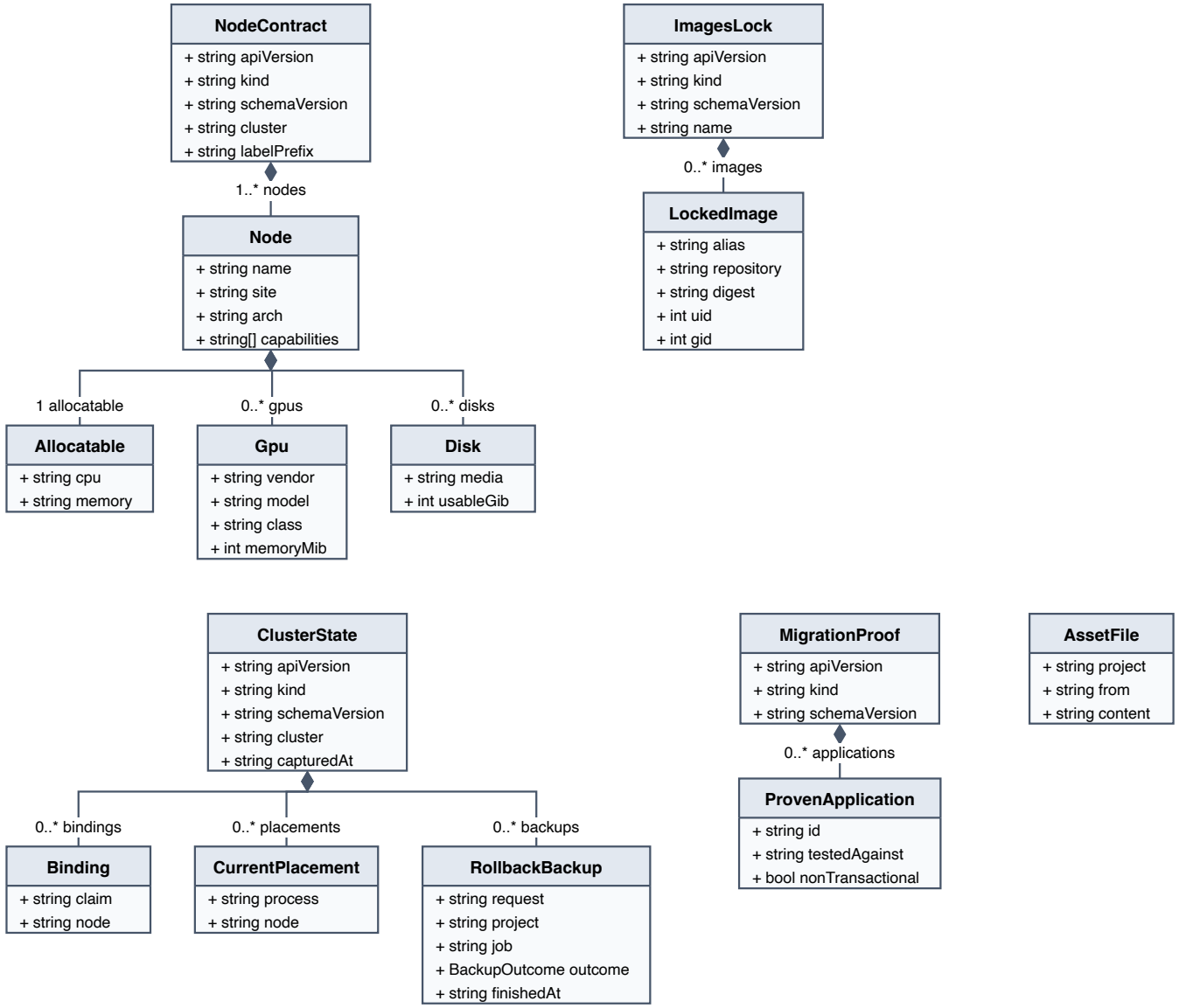



Figure 3: The Pinned Inputs Metamodel. All classes are new in Task 2. Diamonds mark containment.

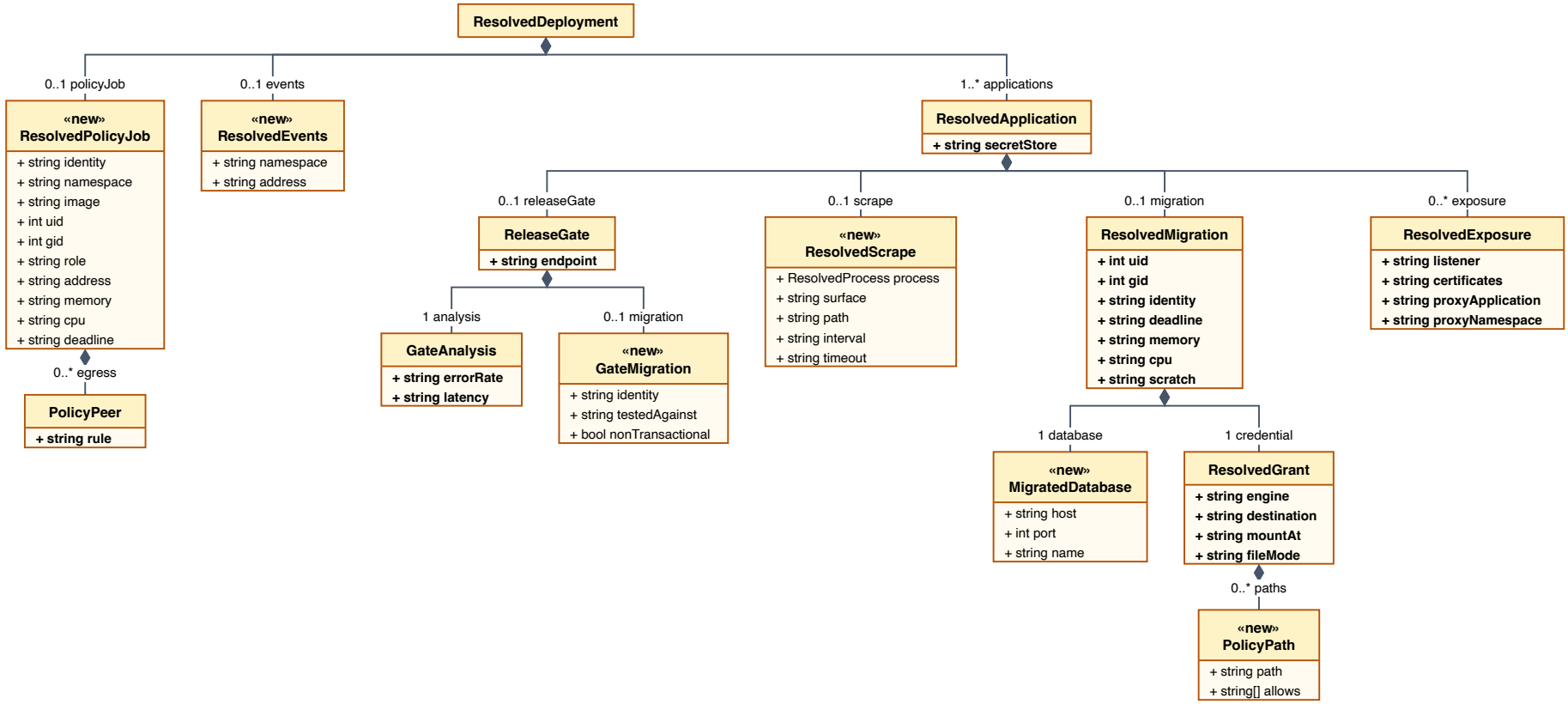



Figure 4: Changed Deployment and Application Classes of the Target Metamodel. A class marked «new» is new in Task 2. A Task 1 class shows only the features Task 2 added, in bold. Diamonds mark containment.

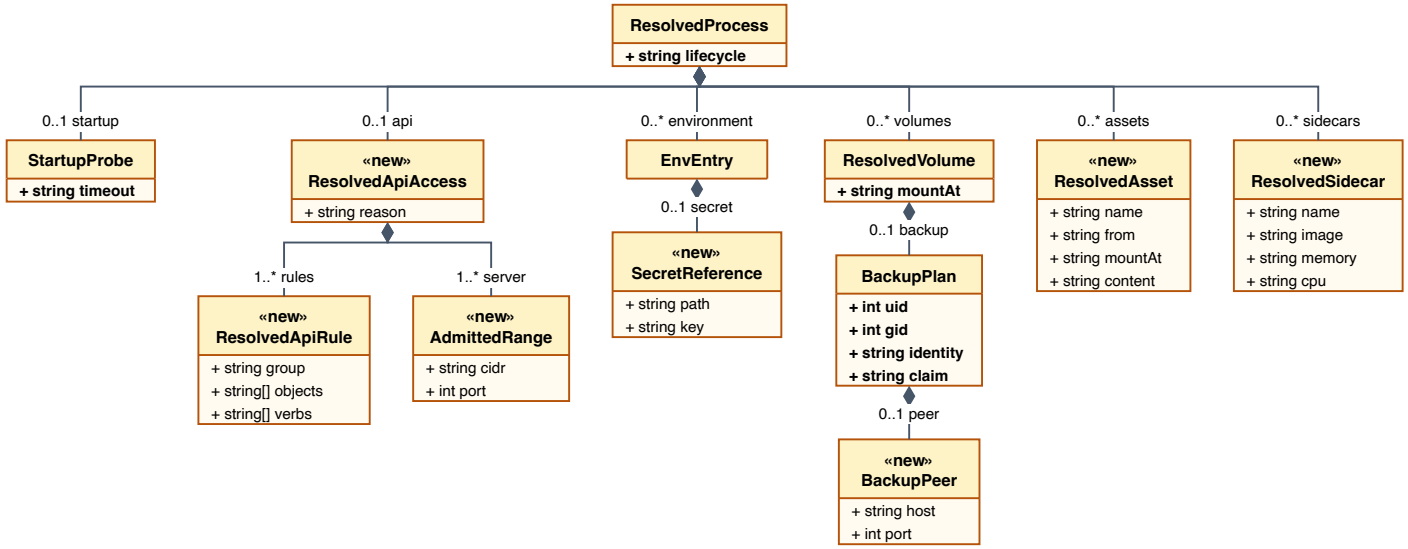



Figure 5: Changed Process Classes of the Target Metamodel, Marked as in Figure 4.

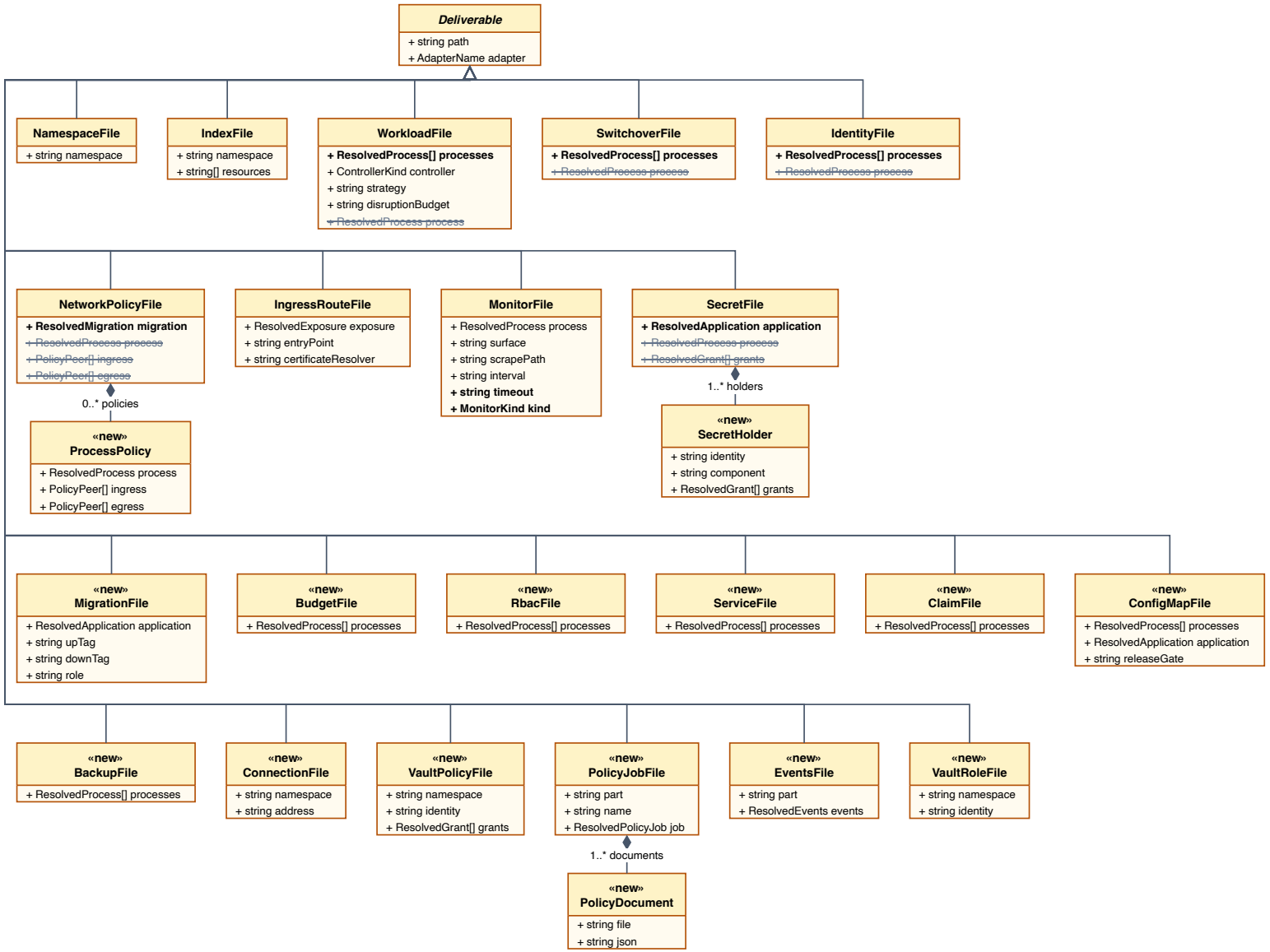



Figure 6: The Generated-File Classes of the Target Metamodel: Task 1's Kinds in the First Two Rows, Task 2's in the Last Two. In a Task 1 class, bold marks an added feature and strikethrough a removed one.